# The Effect of Concentrated Public Housing on Crime in Chicago

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*Original course paper condensed and revised*

**Abstract**

I explore whether increasing public housing in Chicago census tracts correlates with crime changes over time. Using panel data (2009–2013) merging HUD public housing counts with city crime data at 2010 census tracts, I estimate log-linear crime models. The main model includes tract and year fixed effects, so it identifies within-tract changes over time. Results show a positive, significant link between public housing units and crime counts over time. A secondary model with a smaller HUD-assisted sample indicates household and neighborhood composition, like race and economic stability, strongly relate to crime, while the public housing effect is insignificant. These are correlational findings, not causal. Policy implications involve expanding assisted housing and safety strategies without reinforcing concentrated disadvantage.

## 1. Policy Question and Motivation

Cities often debate policies: expanding affordable housing and improving safety. Public housing addresses the first, but concentration can impact neighborhood safety. Chicago's history of public housing, segregation, demolition, and redevelopment illustrates these tensions.

This paper investigates whether increasing public housing in Chicago census tracts correlates with changes in crime over time. While housing debates often focus on neighborhood safety, they are sometimes driven by fear rather than data. Instead of repeating assumptions, I examine what the data shows. This is crucial for city decisions on assisted housing placement, redevelopment, and balancing neighborhood investment with safety.

## 2. Background and Conceptual Framework

Two ideas drive a link between concentrated public housing and crime. First, it can reflect and reinforce poverty; limited access to quality schools and jobs increases crime risk through fewer opportunities, weaker social control, and more violence. Second, concentration influences resource allocation: policing, services, and politics often follow neighborhood stigma and disorder. Crime isn’t inherent to public housing, but it reflects deeper structural disadvantage and institutional responses to it.

At the same time, public housing can lower crime by enhancing housing stability and reducing homelessness, especially when combined with strong property management, resident services, and neighborhood investment. Therefore, the relationship between these factors is not obvious beforehand. Empirically, the goal is to determine whether, after accounting for observable socioeconomic shifts and constant tract features, increases in public housing units within a tract align with changes in crime rates within the same tract.

## 3. Prior Research (selected)

A large body of literature examines the connections among housing policy, neighborhood poverty, and public safety. Three findings are particularly relevant.

### 3.1 Concentration and neighborhood disadvantage

Research on urban inequality shows that concentrated poverty and segregation lead to neighborhood conditions with higher crime. Crime responds nonlinearly to disadvantage, so

small poverty increases can significantly impact already-disadvantaged areas. This indicates that the spatial concentration of assisted housing can intensify disadvantage, not just reflect it.

### 3.2 Demolition/redevelopment and displacement

Chicago studies show that removing distressed public housing can lower local violent crime but has a limited impact on receiving neighborhoods. This suggests policy changes affecting housing density may shift crime spatially rather than reduce overall citywide crime, emphasizing neighborhood-level assessment.

### 3.3 Composition and labor-market attachment

Other research indicates that household composition and economic stability, such as dependence on wages, family structure, and housing mismatches, are linked to neighborhood stability and safety. If these factors differ significantly across assisted housing communities, they could account for the observed links between public housing and crime, even if public housing itself isn't the direct cause.

Together, prior research favors a tract fixed-effects approach: comparing a tract to itself over time reduces bias from stable unobserved traits like long-term segregation and built environment. Still, time-varying omitted variables are a concern and discussed in limitations.

## 4. Data and Sample Construction

### 4.1 Housing data

Housing measures come from HUD's Picture of Subsidized Households, which provides tract-level counts for multiple assisted housing programs. The main independent variable is the number of public housing units in each census tract-year. I restrict analysis to Cook County / Chicago tracts and merge to 2010 tract boundaries for consistency over time.

### 4.2 Crime data

Crime data comes from the City of Chicago's "Crimes 2001–Present" portal. I restrict incidents to 2009–2013, geocode each incident using reported coordinates, and assign incidents to 2010 census tracts using tract boundary files. Incidents are then aggregated to tract-year counts. The main outcome is the natural log of the tract-year incident count (log crime count).

### 4.3 Panel structure and missingness

The merged dataset forms a tract-by-year panel. Some controls, including poverty, minority share, owner share, and income per person, are missing in some tract-years. As a result, fully controlled models use a smaller balanced subset. Throughout, I report the number of observations and emphasize that comparisons across specifications should consider sample changes.

## 5. Measures

Key measures used in the main and secondary specifications include:

- Outcome: Log crime count in tract-year (log of incident count).
- Main predictor: Public housing units in tract-year.
- Socioeconomic controls (main model): minority share, poverty rate, owner share, income per person, plus year fixed effects.
- Secondary (HUD-assisted subsample) controls: female head with children, Black share, Hispanic share, wage as primary income share, % of local median income, and overhoused (housing mismatch).

Using a log crime outcome reduces sensitivity to extreme values and allows coefficients to be interpreted approximately as percent changes. Because public housing units are measured as counts, effect sizes are most meaningful when interpreted for realistic changes (e.g., +50 or +100 units), not one-unit changes.

## 6. Empirical Strategy

### 6.1 Main fixed-effects model

The main model estimates:

$$\ln(\text{Crime}_{it}) = \beta \cdot \text{PublicHousingUnits}_{it} + X_{it} \cdot \theta + \alpha_i + \gamma_t + \varepsilon_{it}$$

where i indexes tracts and t indexes years. $\alpha_i$ are tract fixed effects (absorbing time-invariant tract characteristics such as geography, long-run housing stock, and baseline segregation), and $\gamma_t$ are year fixed effects (capturing citywide shocks such as macroeconomic change or broad policing shifts). $X_{it}$ includes time-varying socioeconomic controls. Standard errors are interpreted as robust to heteroskedasticity and clustered at the tract level.

### 6.2 Identification intuition

Conceptually, tract fixed effects compare a tract to itself over time. If a tract's public housing units increase, $\beta$ measures whether crime in that tract rises or falls during the same period, controlling for observed factors and citywide shocks. This approach improves on cross-sectional models by eliminating stable neighborhood differences, like distance to downtown or long-term disadvantage.

### 6.3 Secondary model (HUD-assisted subsample)

To explore compositional correlates among tracts with assisted housing, I estimate an OLS specification with year fixed effects on a smaller HUD-assisted subsample:

$$\ln(\text{Crime}_{it}) = \beta \cdot \text{PublicHousingUnits}_{it} + X_{it} \cdot \theta + \gamma_t + \varepsilon_{it}$$

This model is not intended as a stronger causal design; rather, it assesses whether demographic and economic measures among assisted households are associated with crime patterns in these tracts.

## 7. Results

### 7.1 Main fixed-effects results

Table 1 reports the headline estimates from the original analysis. Column (3) is the preferred fixed-effects specification, including tract fixed effects, year fixed effects, and socioeconomic controls. The coefficient on public housing units is positive and statistically significant in the preferred model.

| Dependent variable: Log Crime Count | Model (1) | Model (2) | Model (3) Preferred |
|---|---|---|---|
| Public housing units | 0.00035* | 0.00047*** | 0.0018*** |
| Minority share | - | - | 0.225*** |
| Owner share | - | - | -0.228* |
| Poverty % | - | - | 0.025 |
| Income per person | - | - | 0.000 |
| Tract fixed effects | No | No | Yes |
| Year fixed effects | No | Yes | Yes |

| | | | |
|---|---|---|---|
| Observations | 1,018 | 1,018 | 1,018 |
| R-squared | 0.001 | 0.011 | 0.485 |
| Notes | Robust SE; * p<.1, ** p<.05, *** p<.01 | | |

Interpretation: Because the dependent variable is in logs, coefficients can be read as approximate percent changes in crime. The preferred coefficient (0.0018) implies that a one-unit increase in public housing units is associated with roughly a 0.18% increase in crime. More realistically, an increase of 50 units would be roughly 9%, and 100 units about 18%, assuming linearity and controls. These should be interpreted cautiously because the design is correlational and changes in public housing may correlate with other neighborhood changes.

### 7.2 Secondary results: HUD-assisted subsample

Table 2 summarizes the preferred column from the secondary analysis. In this smaller sample, the public housing coefficient is not statistically significant, while several composition variables are strongly associated with crime. Higher Black and Hispanic shares among HUD-assisted households are positively associated with crime counts, while wage as primary income is negatively associated, consistent with a role for economic stability.

| Dependent variable: Log Crime Count (HUD-assisted subsample) | Preferred (Col 4) |
|---|---|
| Public housing units | 0.000408 (0.00029) |
| Single-family owners (%) | -0.013** (0.006) |
| Female head w/ children (%) | -0.0003 (0.00236) |
| Black (%) | 0.013*** (0.002) |
| Hispanic (%) | 0.007** (0.003) |
| Wage as primary income (%) | -0.011*** (as reported) |
| Overhoused (%) | 0.011*** (0.004) |
| Observations | 256 |
| Adj. R-squared | 0.1917 |
| Notes | From original Table 3; * p<.1, ** p<.05, *** p<.01 |

### 7.3 What the two sets of results suggest

The fixed-effects model links increased public housing within-tracts to higher crime. The subsample finds economic and compositional stability strongly relate to crime in assisted-housing tracts, making the public housing coefficient less precise. One view is that public housing concentration proxies for disadvantage and institutional factors associated with crime. Another is that the smaller subsample reduces statistical power, leading to less precise estimates. Overall, the results suggest that public housing's relation to crime depends more on surrounding factors than on public housing alone.

## 8. Robustness, Threats to Validity, and Alternative Explanations

Even with tract- and year-fixed effects, several threats to validity remain.

- Time-varying omitted variables: Changes in policing, gang activity, economic shocks, or redevelopment could impact crime and alter assisted housing counts.
- Reverse causality/policy response: If crime trends influence housing policy (e.g., targeted redevelopment or siting decisions), β may partly reflect policy responses to crime.
- Measurement and reporting: Crime data show reported incidents; reporting or enforcement changes can alter measured incidents regardless of actual crime.
- Functional form: The relationship may be nonlinear (e.g., threshold effects once concentration passes certain levels). A linear count-based predictor may average over different regimes.

These limitations matter for understanding associations. Next, to strengthen causal inference in policy, use quasi-experimental methods like changes in public housing supply, court-ordered relocations, phased rollouts, or difference-in-differences with credible comparison areas.

## 9. Policy Implications for Chicago and Similar Cities

Even though these results are not causal, they still offer a useful lesson for how housing programs are designed and evaluated.

### 9.1 Avoid recreating concentrated disadvantage

If increases in public housing units are associated with increases in crime within the same tract, a cautious policy response is to avoid siting decisions that further concentrate assisted units in already-disadvantaged tracts. Where feasible, cities can pursue mixed-income redevelopment, scattered-site units, and voucher-based strategies paired with strong mobility supports to reduce spatial concentration.

### 9.2 Pair housing investments with safety and opportunity investments

The secondary results show that measures related to economic stability, such as wage-based primary income and housing mismatch, are linked to crime in assisted-housing areas. These links suggest that place-based safety strategies may be more effective when they address household stability and opportunity. This could mean that housing investments work better when supported by other policies, such as workforce programs, youth employment, neighborhood improvement, and community services.

### 9.3 Build evaluation capacity into housing expansions

Because debates around public housing are politically tense, cities should be open about how they evaluate the neighborhood effects of new developments and redevelopment efforts. Agencies can routinely integrate housing program data with safety outcomes to monitor neighborhood impacts. When new units are added or redeveloped, agencies can pre-register evaluation questions, define comparison geographies, and track outcomes over multi-year horizons to distinguish short-term displacement effects from long-term neighborhood change.

## 10. Conclusion

This sample used tract-level data to see if changes in public housing concentration are linked to crime in Chicago census tracts. The preferred model finds a positive, significant link between public housing units and log crime counts. A smaller analysis shows household composition and economic stability among assisted households are strongly tied to crime outcomes. Since the analysis is correlational, we can note associations but not causation. The key policy lesson: housing, safety, and neighborhood conditions are interconnected; improving one requires improving the others.

## Selected References


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